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THE WHIRLPOOL GALAXY

1 Grand spiral

Long lines of stars and dust-laced gas wind round the center of the M51, known as the Whirlpool Galaxy. The arms are star-forming factories where hydrogen gas is compressed and new stars are born. The young hot stars make the arms look bluish, and cause clouds of hydrogen to glow pink. The small galaxy (NGC 5195) at right is passing behind the Whirlpool, triggering star formation as it glides by.

2 Galaxy core

When seen imaged in X-rays, the galaxy's core shines brightly. This image, taken by the Chandra X-ray Observatory, reveals vast clouds of multi-million degree gas at either side of it. The cloud at upper left of the bright central core is 1,500 light-years across. The gas is heated by a high-velocity jet of material accelerating away from a supermassive black hole within the galaxy's nucleus.

3 Inside the core

This Hubble Space Telescope image takes us to the very heart of the galaxy—the active galactic nucleus (AGN) of its central core. The dark "X" silhouetted against the bright nucleus marks the exact location of a black hole, but hides it and its disk of infalling hot gas from view. The broad line of the X is a dust ring 100 light-years across lying at right angles to the galaxy's disk.

4 X-ray view

More than 400 X-ray sources are revealed in this image of the Whirlpool by the space-based Chandra X-ray Observatory, which took 11 hours of observation to create. Most are X-ray binary star systems in which a neutron star, or more rarely a stellar black hole, captures material from an orbiting companion star. The infalling material heats to millions of degrees, producing a luminous X-ray source.